

ST6101: Advanced Statistical Theory (I)

Lecture 7: UMVUE and C-R Lower Bound

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*Adapted from Dr. Dongming Huang's slides.

Example: Uniform Distributions

Let $X_1, \dots, X_n$ be i.i.d. from the uniform distribution on the interval $(0, \theta)$. Suppose now that $\Theta = [1, \infty)$.

- The UMVUE of θ is

$$h(X_{(n)}) = \begin{cases} 1 & 0 \leq X_{(n)} \leq 1 \\ (1 + n^{-1})X_{(n)} & X_{(n)} > 1. \end{cases}$$

- In fact, $h(X_{(n)})$ is complete and sufficient for $\theta \in [1, \infty)$
- Define $T(X) = \max(1, X_{(n)})$, then $h(X_{(n)}) = I_{T=1} + (1 + n^{-1})TI_{T>1}$, i.e., a function of T

Example (Cont.)

We can show that $\max(X_{(n)}, 1)$ is complete and sufficient.

Sufficiency: The joint p.d.f. of $X_1, \dots, X_n$ is

$$\frac{1}{\theta^n} I_{(0, \theta)}(X_{(n)}) = \frac{1}{\theta^n} I_{(0, \theta]}(\max(X_{(n)}, 1)), \text{ a.e.}$$

because $\theta \geq 1$

Completeness: Suppose $\mathbb{E}[f(\max(X_{(n)}, 1))] = 0$ for all $\theta \geq 1$, i.e.

$$0 = \int_0^1 f(1)x^{n-1}dx + \int_1^\theta f(x)x^{n-1}dx, \quad \forall \theta \geq 1$$

Setting $\theta = 1$, we obtain that $f(1) = 0$. Then

$$0 = \int_1^\theta f(x)x^{n-1}dx, \quad \forall \theta > 1,$$

which implies $f(x) = 0$ a.e. for $x > 1$.

Example: Uniform Distributions with Fixed Length

Let X be a sample (of size 1) from the uniform distribution $\text{Unif}(\theta - \frac{1}{2}, \theta + \frac{1}{2})$, $\theta \in \mathcal{R}$. Let g be a **non-constant smooth** function. Then there is no UMVUE of $\eta = g(\theta)$.

- To find a UMVUE of $g(\theta)$, we need to characterize the “0-estimators”
- An unbiased estimator $U(X)$ of 0 must satisfy

$$\int_{\theta - \frac{1}{2}}^{\theta + \frac{1}{2}} U(x) dx = 0 \quad \text{for all } \theta \in \mathcal{R}.$$

- Differentiating both sides of the previous equation and applying the result of differentiation of an integral:

$$U(x) = U(x + 1) \quad \text{a.e.}$$

- That is, U is an unbiased estimator of 0 $\Leftrightarrow U$ has a period of 1 (a.e.) and $\int_0^1 U(x) dx = 0$ (by setting $\theta = 1/2$)

Example (Cont.)

- T is a UMVUE of $g(\theta) \Leftrightarrow$ for any $U(X)$ unbiased for 0 with finite variance, $T(X)U(X)$ is unbiased for 0
- Using the same argument in the last slide, we have

$$T(x)U(x) = T(x+1)U(x+1) \quad \text{a.e.}$$

- Choosing $U(x) = \sin(2\pi x + a)$ with any $a \in \mathbb{R}$, we conclude

$$T(x) = T(x+1) \quad \text{a.e.}$$

- Since T is unbiased for $g(\theta)$,

$$g(\theta) = \int_{\theta - \frac{1}{2}}^{\theta + \frac{1}{2}} T(x) dx \quad \text{for all } \theta \in \mathcal{R}$$

- Differentiating both sides and applying the result of differentiation of an integral

$$g'(\theta) = T\left(\theta + \frac{1}{2}\right) - T\left(\theta - \frac{1}{2}\right) \quad \text{a.e.}$$

- But RHS=0, a.e., which forces g to be a constant function

Lower Bound

- It is often possible to derive the **Cramér-Rao Lower Bound** for the variance of an unbiased estimator
- Idea:
 - For many models, we can define a quantity $I(\theta)$
 - Any unbiased estimator for θ will have variance no less than $1/I(\theta)$
- Let's define such a quantity

Fisher information

- Suppose $\mathcal{P} = \{f_\theta : \theta \in \Theta\}$ where $f_\theta(x)$ is a p.d.f. with parameter θ w.r.t. ν and Θ is an open subset of $\mathbb{R}$
- Suppose for any $\theta \in \Theta$, $\frac{\partial f_\theta(x)}{\partial \theta}$ exists and is finite, P_θ -a.s.
- Let X be a sample from $P_\theta \in \mathcal{P}$
- To measure the amount of information that an observation X carries about θ , we look at the *Fisher information* defined as

$$\begin{aligned} I(\theta) &= \mathbb{E} \left(\frac{\partial}{\partial \theta} \log f_\theta(X) \right)^2 \\ &= \int \left(\frac{\partial}{\partial \theta} \log f_\theta(X) \right)^2 f_\theta(X) d\nu(x). \end{aligned}$$

- We will see soon that if $I(\theta)$ is large, θ can be accurately estimated.

Example: Poisson Families

Suppose $(X_1, \dots, X_n)$ is a i.i.d. sample from a Poisson distribution $\mathcal{P}(\lambda)$. Then

- The joint PMF is

$$f_{\lambda}(x) = \prod_{i=1}^n \frac{\lambda^{x_i}}{x_i!} \exp(-\lambda)$$

- $\log f_{\lambda}(x) = \sum_i x_i \log(\lambda) - n\lambda - \sum_i \log(x_i!)$
- $\frac{\partial}{\partial \lambda} \log f_{\lambda}(x) = \frac{\sum_{i=1}^n x_i}{\lambda} - n$
- $I(\lambda) = \text{Var}\left(\frac{\sum_{i=1}^n X_i}{\lambda}\right) = \frac{n\lambda}{\lambda^2} = \frac{n}{\lambda}$

Here we've used $E \frac{\partial}{\partial \lambda} \log f_{\lambda}(X) = 0$ and thus $I(\lambda) = \text{Var}(\frac{\partial}{\partial \lambda} \log f_{\lambda}(X))$

Example: Normal Families with Known Variance

Let $X_1, \dots, X_n$ be i.i.d. from the $N(\mu, \sigma^2)$ distribution with an unknown $\mu \in \mathcal{R}$ and a known σ^2 .

- The joint Lebesgue p.d.f. is

$$f_{\mu}(x) = \frac{1}{(2\pi\sigma^2)^{n/2}} \exp\left(-\frac{1}{2\sigma^2} \sum_i (x_i - \mu)^2\right)$$

- Then

$$\frac{\partial}{\partial \mu} \log f_{\mu}(x) = \sum_{i=1}^n (x_i - \mu) / \sigma^2$$

- $I(\mu) = \text{Var}\left(\frac{\sum_{i=1}^n (X_i - \mu)}{\sigma^2}\right) = n\sigma^2 / \sigma^4 = n / \sigma^2.$

Property of Fisher Information

- ① $I(\theta)$ depends on the particular parameterization:
- If $\theta = \psi(\eta)$ and ψ is differentiable, then the Fisher information about η is

$$\tilde{I}(\eta) = \psi'(\eta)^2 I(\psi(\eta)),$$

where $I(\theta)$ is the Fisher information about θ .

- ② Suppose that f_θ is twice differentiable in θ and that

$$\int \frac{\partial^2}{\partial \theta^2} f_\theta(x) I_{f_\theta(x) > 0} d\nu = 0$$

Then $I(\theta) = -E \left[\frac{\partial^2}{\partial \theta^2} \log f_\theta(X) \right]$

- ③ Regularity condition (*): $\int \frac{\partial}{\partial \theta} f_\theta(x) d\nu = 0$
- ④ Let X and Y be independent with the Fisher information about θ $I_X(\theta)$ and $I_Y(\theta)$, respectively. Suppose (*) holds for both density functions. Then, the Fisher information about θ contained in (X, Y) is $I_X(\theta) + I_Y(\theta)$.
- Fisher information about θ contained in i.i.d. sample $(X_1, \dots, X_n)$ is $nI_1(\theta)$ where $I_1(\theta)$ is the Fisher information about θ contained in a single X_i provided that (*) holds.

Cramér-Rao Lower Bound

Theorem. Suppose $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$ satisfies the following conditions

- Θ is an open set in $\mathcal{R}$; P_θ has a p.d.f. f_θ w.r.t. a measure ν for all $\theta \in \Theta$
- f_θ is differentiable as a function of θ and satisfies

$$0 = \frac{\partial}{\partial \theta} \int f_\theta(x) d\nu = \int \frac{\partial}{\partial \theta} f_\theta(x) d\nu, \quad \theta \in \Theta \quad (1)$$

Suppose that $g(\theta)$ is a differentiable function.

Let X be a sample from $P \in \mathcal{P}$. Suppose $T(X)$ is an unbiased estimator of $g(\theta)$ such that

$$g'(\theta) = \frac{\partial}{\partial \theta} \int T(x) f_\theta(x) d\nu = \int T(x) \frac{\partial}{\partial \theta} f_\theta(x) d\nu, \quad \theta \in \Theta \quad (2)$$

Then $\text{Var}(T(X)) \geq \frac{g'(\theta)^2}{I(\theta)}$ where $I(\theta) > 0$ for any $\theta \in \Theta$

Proof:

- By the covariance inequality, we have

$$\text{Cov}\left(T(X), \frac{\partial}{\partial \theta} \log f_{\theta}(X)\right)^2 \leq \text{Var}[T(X)] \text{Var}\left[\frac{\partial}{\partial \theta} \log f_{\theta}(X)\right]$$

- By Eq (1), $E \frac{\partial}{\partial \theta} \log f_{\theta}(X) = 0$. Thus
 $\text{Var}\left[\frac{\partial}{\partial \theta} \log f_{\theta}(X)\right] = E\left[\left(\frac{\partial}{\partial \theta} \log f_{\theta}(X)\right)^2\right] = I(\theta)$
- $\text{Cov}(T, \frac{\partial}{\partial \theta} \log f_{\theta}(X)) = E\left[T \frac{\partial}{\partial \theta} \log f_{\theta}(X)\right] = \int T(x) \frac{\partial}{\partial \theta} f_{\theta}(x) d\nu$
- By Eq (2), the last display $= \frac{\partial}{\partial \theta} E[T] = g'(\theta)$
- We conclude that

$$g'(\theta)^2 \leq \text{Var}(T)I(\theta)$$

Remark:

- Equations (1) and (2) are the regularity conditions for the results in Cramér-Rao lower bound and has to be checked
- Typically, they do not hold if the set $\{x : f_{\theta}(x) > 0\}$ depends on θ

Remarks on C-R Lower Bound

- The theorem is also known as *the Information Inequality*.
- The Cramér-Rao lower bound is not affected by any smooth one-to-one reparameterization.
- If an unbiased estimator $T(X)$ of $g(\theta)$ achieves the C-R lower bound, then it is a UMVUE.
 - However, this is not an effective way to find a UMVUE because the Cramér-Rao lower bound is typically *not sharp*.
- Under some regularity conditions, we can show that (left for exercise) there exists an estimator $T(X)$ that attains the C-R lower bound for all $\theta \Leftrightarrow f_\theta$ is of the form $\exp[\eta(\theta)^\top T(x) - \xi(\theta)]h(x)$

Example: Normal Families

Let $X_1, \dots, X_n$ be i.i.d. from the $N(\mu, \sigma^2)$ distribution with an unknown $\mu \in \mathcal{R}$ and a known σ^2 .

- We have showed $I(\mu) = n/\sigma^2$.
- Consider the estimation of μ .
- Note that the sample mean $\bar{X}$ is unbiased and has variance σ^2/n .
- So $\bar{X}$ attains the Cramér-Rao lower bound and it is a UMVUE.

Example: Normal Families (Cont.)

Model

$X_1, \dots, X_n$ be i.i.d. from the $N(\mu, \sigma^2)$ distribution with an unknown $\mu \in \mathcal{R}$ and a known σ^2 . $I(\mu) = n/\sigma^2$.

- Consider now the estimation of $\eta = \mu^2$.
- Since $\mathbb{E}(\bar{X}^2) = \mu^2 + \sigma^2/n$ and $\bar{X}$ is sufficient and complete, the UMVUE of η is $h(\bar{X}) = \bar{X}^2 - \sigma^2/n$ (by Lehmann-Scheffé Theorem).
- A straightforward calculation shows that (left for exercise)

$$\text{Var}(h(\bar{X})) = \frac{4\mu^2\sigma^2}{n} + \frac{2\sigma^4}{n^2}.$$

- Since $g'(\mu) = 2\mu$, the Cramér-Rao lower bound is $4\mu^2\sigma^2/n$.
- Hence $\text{Var}(h(\bar{X}))$ does not attain the Cramér-Rao lower bound.

Extension to Multi-parameter Case

Let X be a sample from $P \in \mathcal{P} = \{f_\theta(x) : \theta \in \Theta\}$, where Θ is an open set in $\mathcal{R}^k$ and the p.d.f. f_θ is differentiable as a function of θ .

- Define the score function

$$\frac{\partial}{\partial \theta} \log f_\theta(X) = \left(\frac{\partial}{\partial \theta_1} \log f_\theta(X), \dots, \frac{\partial}{\partial \theta_k} \log f_\theta(X) \right)^\top$$

- Define the *Fisher information matrix* as the $k \times k$ matrix

$$I(\theta) = \mathbb{E} \left\{ \frac{\partial}{\partial \theta} \log f_\theta(X) \left[\frac{\partial}{\partial \theta} \log f_\theta(X) \right]^\top \right\}$$

- Suppose that f_θ is twice differentiable in θ and that

$$0 = \frac{\partial}{\partial \theta} \int \frac{\partial}{\partial \theta^\tau} f_\theta(x) d\nu = \int \frac{\partial^2}{\partial \theta \partial \theta^\tau} f_\theta(x) d\nu, \quad \theta \in \Theta.$$

Then

$$I(\theta) = -E \left[\frac{\partial^2}{\partial \theta \partial \theta^\tau} \log f_\theta(X) \right]$$

Multivariate C-R Lower Bound (Theorem 3.3)

When θ is k -dimensional, $g : \Theta \mapsto \mathbb{R}$, the inequality in the Cramér-Rao Lower Bound for $ET = g(\theta)$ ($\theta \in \Theta$)

$$\text{Var}(T(X)) \geq \left[\frac{\partial}{\partial \theta} g(\theta) \right]^\top [I(\theta)]^{-1} \frac{\partial}{\partial \theta} g(\theta),$$

where the gradient $\frac{\partial}{\partial \theta} g(\theta) = (\frac{\partial}{\partial \theta_1} g(\theta), \dots, \frac{\partial}{\partial \theta_k} g(\theta))^\top$ is assumed to satisfy the regularity condition:

$$\frac{\partial}{\partial \theta} g(\theta) = \int T(x) \frac{\partial}{\partial \theta} f_\theta(x) d\nu.$$

Here we also assume $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$ satisfies the following conditions:

- Θ is an open set in $\mathcal{R}^k$; P_θ has a p.d.f. f_θ w.r.t. a measure ν for all $\theta \in \Theta$
- f_θ is differentiable as a function of θ and satisfies

$$0 = \frac{\partial}{\partial \theta} \int f_\theta(x) d\nu = \int \frac{\partial}{\partial \theta} f_\theta(x) d\nu, \quad \theta \in \Theta.$$

Sketch of proof

- By the covariance inequality, for any $\mathbf{c} \in \mathbb{R}^k$,

$$\text{Var}(T) \text{Var}\left(\mathbf{c}^\top \frac{\partial \log f_\theta(X)}{\partial \theta}\right) \geq \left[\text{Cov}\left(T, \mathbf{c}^\top \frac{\partial \log f_\theta(X)}{\partial \theta}\right) \right]^2.$$

- Use $\mathbf{a} = \text{Cov}\left(\frac{\partial}{\partial \theta} \log f_\theta(X), T(X)\right)$ to simplify notations
- The LHS is $\text{Var}(T) (\mathbf{c}^\top I(\theta) \mathbf{c})$
- The RHS is $(\mathbf{c}^\top \mathbf{a})^2$
- Choose $\mathbf{c} = [I(\theta)]^{-1} \mathbf{a}$
- Use the regularity condition, we can show $\mathbf{a} = \frac{\partial}{\partial \theta} g(\theta)$

Example: Normal Families

Let $X_1, \dots, X_n$ be i.i.d. $\sim N(\mu, \nu)$. Let $\theta = (\mu, \nu)$. Then

$$\log f_{\theta}(\mathbf{x}) = -\frac{1}{2\nu} \sum_{i=1}^n (x_i - \mu)^2 - \frac{n}{2} \log(2\pi\nu).$$

It can be calculated that

$$\begin{aligned}\frac{\partial^2}{\partial \mu^2} \log f_{\theta}(\mathbf{x}) &= -\frac{n}{\nu}, \\ \frac{\partial^2}{\partial \nu^2} \log f_{\theta}(\mathbf{x}) &= -\frac{\sum_{i=1}^n (x_i - \mu)^2}{\nu^3} + \frac{n}{2\nu^2}, \\ \frac{\partial^2}{\partial \nu \partial \mu} \log f_{\theta}(\mathbf{x}) &= -\frac{\sum_{i=1}^n (x_i - \mu)}{\nu^2}.\end{aligned}$$

Thus, the Fisher information matrix about θ contained in $X_1, \dots, X_n$ is

$$I(\theta) = -E \left[\frac{\partial^2}{\partial \theta \partial \theta^{\tau}} \log f_{\theta}(X) \right] = \begin{pmatrix} \frac{n}{\nu} & 0 \\ 0 & \frac{n}{2\nu^2} \end{pmatrix}.$$

Exercise

Let $X_1, \dots, X_n$ be i.i.d. $\sim N(\mu, \nu)$. Let $\theta = (\mu, \nu)$.
Find the C-R lower bound for $\mu^2 - 2\nu$.

Proposition

Suppose that the distribution of X is from an exponential family $\{f_\theta : \theta \in \Theta\}$, i.e., the p.d.f. of X w.r.t. a σ -finite measure is

$$f_\theta(x) = \exp\{[\eta(\theta)]^\top T(x) - \xi(\theta)\} h(x), \quad (3)$$

where Θ is an open subset of $\mathcal{R}^k$.

(i) For any S with $\mathbb{E}|S(X)| < \infty$, it holds that

$$\frac{\partial}{\partial \theta} \int S(x) f_\theta(x) d\nu = \int S(x) \frac{\partial}{\partial \theta} f_\theta(x) d\nu, \quad \theta \in \Theta$$

and

$$I(\theta) = -\mathbb{E} \left[\frac{\partial^2}{\partial \theta \partial \theta^\top} \log f_\theta(X) \right].$$

This is a direct consequence of Theorem 2.1 (of the textbook).

Proposition (Cont.)

- (ii) If $\underline{I}(\eta)$ is the Fisher information matrix for the natural parameter η , then the covariance matrix $\text{Var}(T) = \underline{I}(\eta)$.

Proof:

- The p.d.f. under the natural parameter η is

$$f_{\eta}(x) = \exp \left\{ \eta^{\top} T(x) - \zeta(\eta) \right\} h(x).$$

From Theorem 2.1 of (the textbook), $\mathbb{E}[T(X)] = \frac{\partial}{\partial \eta} \zeta(\eta)$. The result follows from

$$\frac{\partial}{\partial \eta} \log f_{\eta}(x) = T(x) - \frac{\partial}{\partial \eta} \zeta(\eta).$$

Proposition (Cont.)

(iii) Let $\psi = \mathbb{E}[T(X)]$. Suppose $\bar{I}(\psi)$ is the Fisher information matrix for the parameter ψ , then $\text{Var}(T) = [\bar{I}(\psi)]^{-1}$.

- Since $\psi = \mathbb{E}[T(X)] = \frac{\partial}{\partial \eta} \zeta(\eta)$,

$$\underline{I}(\eta) = \frac{\partial \psi^\top}{\partial \eta} \bar{I}(\psi) \left(\frac{\partial \psi^\top}{\partial \eta} \right)^\top = \frac{\partial^2}{\partial \eta \partial \eta^\top} \zeta(\eta) \bar{I}(\psi) \left[\frac{\partial^2}{\partial \eta \partial \eta^\top} \zeta(\eta) \right]^\top.$$

- By Theorem 2.1 (of the textbook) (see also Exercise 2 in Tutorial 4) and the result in (ii),

$$\frac{\partial^2}{\partial \eta \partial \eta^\top} \zeta(\eta) = \text{Var}(T) = \underline{I}(\eta).$$

- Hence

$$\bar{I}(\psi) = [\underline{I}(\eta)]^{-1} \underline{I}(\eta) [\underline{I}(\eta)]^{-1} = [\underline{I}(\eta)]^{-1} = [\text{Var}(T)]^{-1}.$$